JAIN FACULTY OF ENGINEERING AND TECHNOLOGY

Department of Computer Science & Engineering

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PROJECT CENTRIC LEARNING (PCL) PHASE 2

PROJECT TEAM NO.	CoEPCL-0213[41]
PROJECT TITLE	ZK Credential Verification with Blockchain and LLM Parsing

TEAM MEMBERS

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DETAILS OF THE PROJECT GUIDE

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DOMAIN OF THE PROJECT

The domain of the project include Cyber Security, Blockchain Technology, Artificial Intelligence and (ZKP) Zero - knowledge proof Systems.

Scope of the Project

1. Secure User Authentication and Encrypted document Management
2. AI - Based OCR, Claim Extraction, and Validation.
3. Zero - knowledge Proof Generation with Blockchain Verification
4. Scalable Backend Processing and Frontend Dashboard Deployment.

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PROJECT GUIDE

PROJECT COORDINATOR

Existing System:-

The existing credential verification systems are mainly document-centric and disclosure-heavy systems. Users are required to upload full documents for verification purpose.

Features:-

* Manual or semi-automatic document verification.
* Full document sharing
* Centralized storage of sensitive data
* Rule based extraction systems

Drabacks of Existing Systems:-

* Exposure of personal information.
* Risk of data breaches
* Unauthorized access.
* High dependency on centralized trust
* Limited privacy protection
* High storage and compilance burden

Problem Identification

The problem identified are:-

1. Privacy Leakage
2. Data breach risks
3. Trust Fragmentation
4. Document Hetrogenety
5. Lack of Cryptographic Assurance
6. Compilance challenges.

Proposed System:-

The proposed system is a Zk Credential Verification system using blockchain and LLM parsing that enables privacy-preserving credential verification using AI and cryptography.

Features of Proposed System:-

1. LLM based document parsing.
2. OCR based processing
3. Zero Knowledge proof generation
4. Blockchain Verification
5. Minimal Data Disclosure
6. Secure Storage
7. Asynchronous scalable processing
8. Trustless verification

No. of Modules

Sl. No.	Name of the Module	Remarks
1.	User Authentication and Security	
2.	Document Upload and storage Module	
3.	OCR and Document processing module.	
4.	LLM - Based claim Extraction Module.	
5.	Claim validation and Normalization Module	
6.	Blockchain and smart contract module.	

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